



CODE SPORT

In this month's column, we continue our discussion on 'data science' and focus on the topic of information retrieval.

Data science is a multi-disciplinary subject requiring knowledge in machine learning, information retrieval, statistics, data mining, etc. We start our journey on data science by focusing on information retrieval.

What is information retrieval?

Here's the textbook definition of information retrieval (IR), as defined by the IR books: "Information retrieval (IR) is finding material (usually documents) of an unstructured nature (usually text) that satisfies an information need from within large collections (usually stored on computers)." This definition can be extended to cover multimedia content and images as well, in today's context, apart from just text. While a simple example of information retrieval would be to look up a phone number in your phone book, the most well-known example of information retrieval we all encounter often is the Web search.

For instance, let us assume that we are looking for 'Italian restaurants which serve vegetarian food in Bangalore'. So we would type the following query as a set of keywords into Google's search bar, 'Italian restaurant vegetarian Bangalore', in response to which we get a set of Web pages. Some of the Web pages returned may be relevant to our query and some may not be very relevant. While our information need actually involved a question, we translated that information need into an approximation, namely, a set of keywords which appear in that question, and used the set of query terms to search for information on the World Wide Web.

Note that approximating our information need into a set of key words can actually lead to degradation in the quality of information results that are returned to us. For instance, the query possibly returned all Italian restaurants that served

both vegetarian and non-vegetarian food. Instead of expressing our information need in terms of a strict Boolean query, we approximated it as a set of key words. The Boolean formulation of the above query can be of the form: "(location == Bangalore) AND (restaurant type == ITALIAN) AND (food type == VEG)". The Boolean query is a much stricter and accurate formulation of our information needs, but its disadvantage is that the user needs to be familiar with the syntax of the Boolean query, unlike the traditional natural language representation of the query by just using the keywords.

Now let us try to understand how the keyword query actually retrieves a set of Web pages as a result to that query. For the sake of simplicity, let us consider the set of Web pages containing details about restaurants as a set of text documents (this is actually an unrealistic assumption given that Web pages are in HTML form and much of the information that is presented in a Web page can be generated on the fly by JavaScript code, which makes it difficult for crawlers to get the complete and static view of the page's content).

We can now restate our problem as follows: Given a set of text documents $D = \{d_j\}$ where j can vary from 1 to N , where N is the total number of documents in the set D . ' j ' represents the unique index by which the document in the collection can be identified. Each document contains a number of words or terms. I use 'terms' and 'words' interchangeably in our discussion. The total number of distinct terms in our document collection is $\{t_1, t_2, \dots, t_k\}$. The set of terms in our collection is also known as the 'vocabulary'. The query ' q ' also consists of a set of words or terms.

Our example query of Italian restaurants consists of the four key words: Italian, restaurant, vegetarian